

CNN Transfer Learning versus a Zero-Shot Vision-Language Model for Bone Fracture Identification in X-Ray Images

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Abstract

Bone fracture identification from radiographs is a clinically important computer vision problem because missed fractures can delay treatment, while false alarms can increase clinical workload. This paper presents a reproducible research prototype for binary fracture classification using FracAtlas, a public musculoskeletal radiograph dataset with image-level fracture labels. A custom convolutional neural network (CNN) baseline was compared with three ImageNet-pretrained transfer-learning models: MobileNetV2, ResNet50, and DenseNet121. The CNN experiment used 2,858 training images, 612 validation images, and 613 held-out test images. Because fracture-positive samples were the minority class, training used weighted binary cross entropy with logits. ResNet50 achieved the strongest CNN test performance, with accuracy of 0.8581, recall of 0.7593, F1-score of 0.6534, and ROC-AUC of 0.8915. To contextualize this result, the paper also discusses the ResNet50 result relative to a recently reported weakly supervised FracAtlas benchmark, while avoiding a strict state-of-the-art claim because data splits, supervision assumptions, preprocessing, and threshold-selection protocols differ across studies. Grad-CAM visualizations were used to inspect model attention and failure cases. In addition, a zero-shot vision-language model (VLM) baseline was evaluated on the same 613-image held-out test set using GPT-4o-mini with three prompt modes: simple, conservative, and sensitivity-oriented. The best VLM prompt by recall was the sensitivity-oriented prompt, which achieved accuracy of 0.8059, recall of 0.3704, F1-score of 0.4020, and ROC-AUC of 0.6357. Under this dataset and prompt protocol, the zero-shot VLM was highly specific but substantially less sensitive than the task-trained ResNet50 model. Overall, within the internal FracAtlas split, CNN transfer learning provided stronger fracture-detection performance than the tested zero-shot VLM baseline, while the VLM experiment showed that prompt wording can shift the sensitivity-specificity trade-off. This project is a research-only benchmark and is not intended for clinical diagnosis.

Keywords: bone fracture detection; X-ray image classification; convolutional neural networks; transfer learning; ResNet50; Grad-CAM; vision-language models; FracAtlas

1 Introduction

Digital radiography is widely used for suspected musculoskeletal fractures because X-ray imaging is fast, inexpensive, and broadly available. However, fracture interpretation can still be difficult when fractures are subtle, anatomy overlaps, or images contain different positioning and quality conditions. Computer vision models may support research on fracture screening, but medical image models require careful evaluation: a high accuracy score alone does not guarantee safe or clinically meaningful behavior.

This project studies binary fracture classification: given one X-ray image, the model predicts whether the image is fracture-positive or no-fracture. The goal is not to build a deployable clinical diagnostic tool. Instead, the goal is to build a reproducible undergraduate research prototype that compares CNN architectures, evaluates error types, and uses Grad-CAM to audit qualitative model attention. After the CNN experiments, this project also evaluates a zero-shot VLM baseline on the same held-out FracAtlas test set. This turns the project from a CNN-only comparison into a dual-track evaluation: task-trained transfer learning versus a general-purpose multimodal model under a fixed research protocol.

The main contributions are: (1) a complete training and evaluation workflow for FracAtlas fracture classification; (2) a comparison of a custom CNN with MobileNetV2, ResNet50, and DenseNet121; (3) evaluation using sensitivity, specificity, F1-score, and ROC-AUC rather than accuracy alone; (4) Grad-CAM analysis of true positive, true negative, false positive, and false negative cases; and (5) a full held-out test-set VLM baseline using standardized prompts to test zero-shot fracture classification on the same 613 test images used for CNN evaluation.

2 Related Work

FracAtlas is a public dataset for fracture classification, localization, and segmentation of musculoskeletal radiographs. It contains 4,083 X-ray images and was annotated with expert review by radiologists and an orthopedist [1]. Other musculoskeletal radiograph datasets, including MURA [2] and GRAZPEDWRI-DX [3], demonstrate the importance of public benchmark datasets for evaluating fracture and abnormality detection models.

Positioning within FracAtlas work. Publicly reported classification benchmarks on FracAtlas remain more limited than benchmarks on larger musculoskeletal datasets. The original FracAtlas paper primarily introduced the dataset and its classification, localization, and segmentation annotations rather than establishing a single standardized classification leaderboard [1]. A recent weakly supervised multiple-instance learning study, FAD-MIL, evaluated FracAtlas after duplicate removal and reported an AUC of 0.833 and an F1-score of 0.541 on its internal test set, with ABMIL reaching an AUC of 0.817 [4]. These results provide a useful reference point for this paper, but the comparison remains indirect because the present study uses a different image-level split, preprocessing pipeline, model class, and fixed-threshold evaluation protocol.

Transfer learning is common in medical image classification because many medical datasets are small compared with natural-image datasets. The ImageNet Large Scale Visual Recognition Challenge helped standardize large-scale visual pretraining and evaluation [5]. ResNet introduced residual connections to improve training of deep networks [6]. DenseNet uses dense connections to encourage feature reuse and gradient flow [7]. MobileNetV2 uses inverted residuals and linear bottlenecks to provide a lightweight architecture [8]. For fracture detection, previous work has shown that deep learning can support radiograph interpretation, but also requires careful validation and clinician-centered use [9], [10]. Transfer learning has also been studied for musculoskeletal fracture classification [11]. For explainability, Grad-CAM produces class-discriminative heatmaps by using gradients flowing into a convolutional layer [12].

Recent multimodal APIs allow general-purpose models to process image inputs. OpenAI describes GPT-4o-mini as a model that accepts text and image inputs and produces text outputs [13]. OpenAI’s image-input documentation describes using vision-capable models for visual analysis, but also warns that such models are not suitable for interpreting specialized medical images or providing medical advice [14]. Therefore, the VLM experiment in this paper is treated only as a research-only benchmark, not as a medical decision system.

3 Materials and Methods

3.1 Dataset and Label Definition

The experiment used FracAtlas. The global binary label was taken from the `fractured` column, where 1 indicates fracture and 0 indicates no fracture. The image filename was read from the `image_id` column. The full dataset split contained 2,858 training images, 612 validation images, and 613 held-out test images. The training set was imbalanced, with 2,356 no-fracture images and 502 fracture-positive images.

The GitHub repository contains code, metadata, split files, representative outputs, and a smaller runnable image subset. The full FracAtlas image folder is larger than what is practical for the project repository, so reproducing the full CNN and VLM experiments requires downloading the complete FracAtlas images and placing them under the expected `data/FracAtlas/images/`

path. For the final VLM run, the 613 test images were also saved into a clean image folder to avoid failures caused by a small number of truncated JPEG files; this preprocessing changed only the image encoding and did not change labels or split membership.

3.2 Code Availability and Reproducibility

The project repository is available at https://github.com/Kenny-W99/Bone_Fracture_ML_Detection. The training and evaluation scripts save metrics, model checkpoints, figures, run metadata, CSV hashes, and environment information. This makes it possible to track which data split, model configuration, and software environment produced each result.

The VLM extension added four scripts: `vlm.baseline.py`, `evaluate_vlm.py`, `make_vlm_subset.py`, and `append_vlm_to_summary.py`. These scripts build a recursive image index, send X-ray images to a selected VLM provider, parse structured JSON predictions, evaluate standard classification metrics, and optionally append VLM results to the experiment summary.

3.3 Preprocessing and Augmentation

All X-ray images were loaded as three-channel RGB images to match ImageNet-pretrained CNN backbones. Although the radiographs are visually grayscale, RGB conversion allows direct use of pretrained torchvision models. Images were padded to preserve aspect ratio and then resized to 224×224 pixels. Aspect-ratio preservation reduces geometric distortion of bone anatomy.

Training augmentation was intentionally conservative. The pipeline used small rotations, small translations, brightness and contrast jitter, and optional horizontal flipping. Vertical flipping was not used because X-rays have meaningful anatomical orientation. Random resized cropping was also avoided because it could remove a subtle fracture region and create misleading labels.

3.4 Binary Classification Formulation

For each input image x_i , a CNN model f_θ outputs one scalar logit z_i :

$$z_i = f_\theta(x_i). \quad (1)$$

The logit is converted into a fracture probability using the sigmoid function:

$$p_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}. \quad (2)$$

During evaluation, images with $p_i \geq 0.5$ were classified as fracture-positive, while images with $p_i < 0.5$ were classified as no-fracture. This formulation treats the task as binary image classification, not fracture localization or segmentation.

The fixed threshold of 0.5 was used as a simple baseline decision rule so that all CNN models could be evaluated under the same setting. However, this choice is a limitation because weighted loss can shift the distribution of model scores, and

a threshold of 0.5 may not be the best operating point for a screening-oriented task. A systematic threshold search on the validation set is therefore left for future work.

3.5 Class Imbalance and Weighted Loss

Because fracture-positive samples were the minority class, the training objective used positive class weighting in binary cross entropy. With $N_{neg} = 2356$ and $N_{pos} = 502$, the positive class weight was:

$$w_p = \frac{N_{neg}}{N_{pos}} = \frac{2356}{502} \approx 4.69. \quad (3)$$

The weighted binary cross-entropy loss can be written as:

$$\mathcal{L} = -w_p y \log(\sigma(z)) - (1 - y) \log(1 - \sigma(z)). \quad (4)$$

This weighting makes fracture-positive errors contribute more strongly to the loss, reducing the tendency of the model to predict only the majority no-fracture class. It does not remove the need to choose an appropriate evaluation threshold, since loss weighting and threshold selection control different parts of the decision pipeline.

3.6 CNN Architectures and Transfer Learning

Four CNN models were evaluated. The custom CNN served as a baseline trained from scratch. MobileNetV2, ResNet50, and DenseNet121 were initialized with ImageNet-pretrained weights. For the transfer-learning models, the original multi-class classification head was replaced with a single-output binary classifier that produced one logit. This allowed all architectures to use the same binary loss function and evaluation metrics.

3.7 Vision-Language Model Baseline

To compare task-specific CNN transfer learning with a general-purpose multimodal model, a zero-shot VLM baseline was added after the CNN experiments. The VLM was not fine-tuned on FracAtlas. Instead, each X-ray image was sent to GPT-4o-mini with a standardized prompt asking the model to return valid JSON containing a binary prediction, a fracture confidence score, and a short reason. The output was parsed into a CSV file and evaluated against the ground-truth fractured label.

Unlike the earlier repository-only pilot, the final VLM experiment used the full 613-image held-out test set, the same test set used for the CNN models. Three prompt modes were tested. The simple prompt asked for a direct fracture/no-fracture classification. The conservative prompt instructed the model to choose no-fracture when visual evidence was insufficient. The sensitivity-oriented prompt instructed the model to prioritize possible fracture evidence. This design tested whether prompt wording changed recall and false-positive behavior. The VLM confidence score should be interpreted cautiously because it is model-generated and not calibrated in the same way as a CNN sigmoid probability.

3.8 Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall/sensitivity, specificity, F1-score, and ROC-AUC. Because missed fractures are important in medical screening, recall was emphasized alongside accuracy. The main metrics were computed as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (5)$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (6)$$

ROC-AUC was included because it evaluates discrimination across thresholds rather than only at the fixed 0.5 decision threshold. For the VLM, ROC-AUC is reported using the returned fracture confidence score, but this score is not assumed to be clinically calibrated.

4 Experimental Setup

All CNN models were trained with AdamW [15] and a learning rate of 10^{-4} . The custom CNN was trained for 10 epochs, while MobileNetV2, ResNet50, and DenseNet121 were trained for 5 epochs each. The transfer-learning models were trained for fewer epochs because they started from ImageNet-pretrained features and tended to converge faster, while the custom CNN was given more optimization time because it was trained from scratch. However, this unequal training schedule is still a potential confounding factor rather than a perfectly compute-matched comparison. The best checkpoint was selected based on validation F1-score. Although recall is clinically important for screening, using recall alone as the checkpoint criterion could select a model that over-predicts fracture-positive cases; F1-score was used as a balancing criterion between precision and recall in this research prototype. Experiments were run in a PyTorch and torchvision environment with hardware acceleration when available. The final CNN metrics were computed only on the held-out test set.

The final VLM evaluation used GPT-4o-mini through the OpenAI image-input API. All three prompt modes were evaluated on the same 613-image held-out test set. The test set contained 108 fracture-positive images and 505 no-fracture images. Each prompt run produced 613 valid predictions and zero failed predictions after the clean JPEG image folder was used.

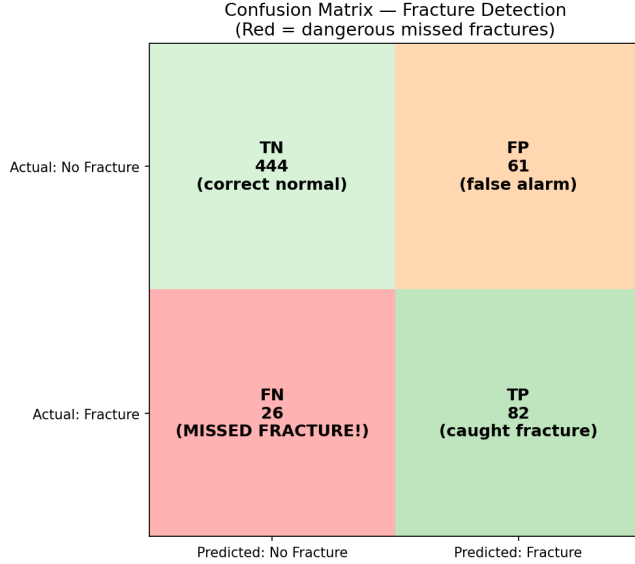
5 Results

5.1 CNN Results

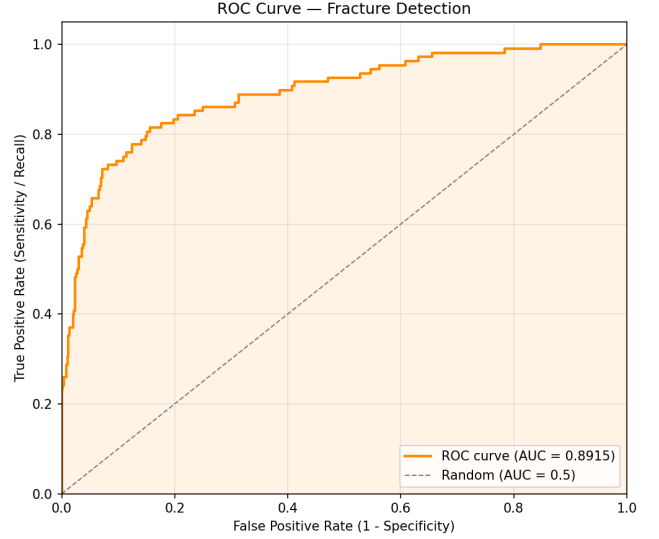
Table 1 summarizes final CNN test performance. On the internal FracAtlas train/validation/test split, the transfer-learning models outperformed the custom CNN baseline. ResNet50 achieved the best accuracy, recall, F1-score, and ROC-AUC, while DenseNet121 achieved the highest specificity but lower recall than ResNet50. The ResNet50 ROC-AUC of 0.8915 and F1-score of 0.6534 are numerically competitive with recently reported FracAtlas results such as FAD-MIL, but this comparison

Table 1: CNN performance on the held-out FracAtlas test set. Recall is emphasized because false negatives correspond to missed fracture-positive images.

Model	Epochs	Accuracy	Precision	Recall	Specificity	F1-score	ROC-AUC	TP	FP	TN	FN
Custom CNN	10	0.7765	0.4027	0.5556	0.8238	0.4669	0.7613	60	89	416	48
MobileNetV2	5	0.8532	0.5662	0.7130	0.8832	0.6311	0.8709	77	59	446	31
ResNet50	5	0.8581	0.5734	0.7593	0.8792	0.6534	0.8915	82	61	444	26
DenseNet121	5	0.8450	0.5520	0.6389	0.8891	0.5923	0.8558	69	56	449	39



(a) ResNet50 confusion matrix



(b) ResNet50 ROC curve

Figure 1: Held-out test evaluation for the best CNN model. ResNet50 detected 82 fracture-positive images and missed 26 fracture-positive images, with ROC-AUC of 0.8915.

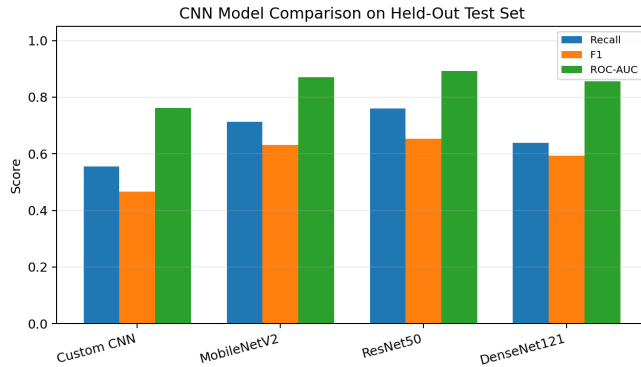


Figure 2: CNN model comparison using recall, F1-score, and ROC-AUC. The pretrained models show stronger dataset-internal performance than the custom CNN baseline.

is only contextual because the studies do not share an identical split or protocol [4]. These findings should be interpreted as dataset-internal performance rather than evidence of external generalization across other hospitals, scanners, body-positioning patterns, or radiographic datasets.

ResNet50 achieved a recall of 0.7593, meaning it correctly identified 82 out of 108 fracture-positive test images. However,

it still produced 26 false negatives. In a screening-oriented task, these missed fractures are more concerning than false positives because they may delay treatment. Therefore, even the best CNN model should be viewed as a research prototype rather than a clinical system.

5.2 VLM Results on the Full Test Set

Table 2 summarizes the GPT-4o-mini zero-shot VLM baseline on the full 613-image held-out test set. The simple prompt achieved high specificity of 0.9822 but low recall of 0.1389, detecting only 15 of 108 fracture-positive images. The conservative prompt was even more cautious, with specificity of 0.9921 but recall of 0.0833. The sensitivity-oriented prompt increased recall to 0.3704 and detected 40 fracture-positive images, but it also increased false positives from 9 to 51 and reduced specificity to 0.8990.

The VLM result changes the paper from a small pilot exploration into a full test-set baseline. Under this specific dataset, prompt design, and GPT-4o-mini setup, the zero-shot VLM showed lower recall, F1-score, and ROC-AUC than the task-trained ResNet50 model. This should not be interpreted as a general claim that all VLMs are inferior to CNNs for medical

Table 2: GPT-4o-mini zero-shot VLM results on the same 613-image held-out FracAtlas test set used for CNN evaluation. The sensitivity-oriented prompt improved recall, while the zero-shot VLM remained conservative under this protocol.

Model	Prompt	n	Valid	Failed	Accuracy	Precision	Recall	Specificity	F1-score	ROC-AUC	TP	FP	TN	FN
GPT-4o-mini	Simple	613	613	0	0.8336	0.6250	0.1389	0.9822	0.2273	0.5702	15	9	496	93
GPT-4o-mini	Conservative	613	613	0	0.8320	0.6923	0.0833	0.9921	0.1488	0.5386	9	4	501	99
GPT-4o-mini	Sensitive	613	613	0	0.8059	0.4396	0.3704	0.8990	0.4020	0.6357	40	51	454	68

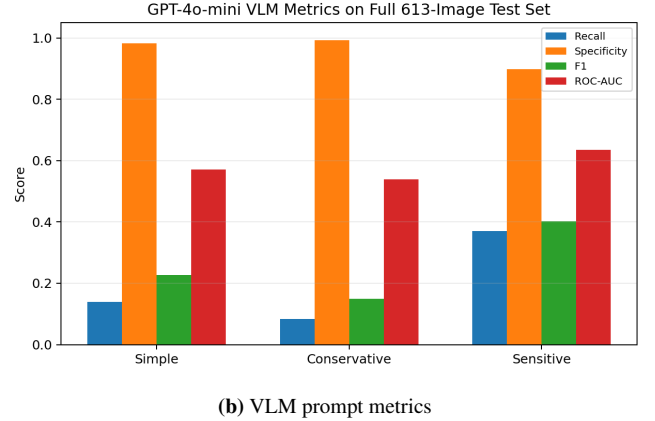
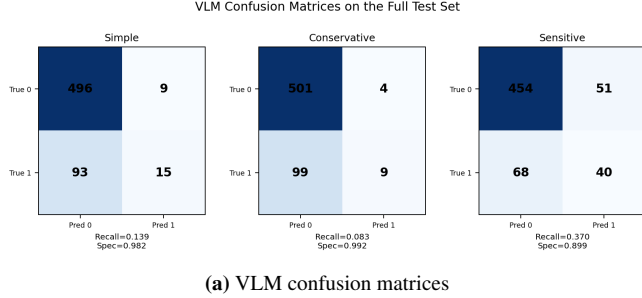


Figure 3: Full test-set VLM evaluation. GPT-4o-mini was generally conservative in fracture-positive prediction. The sensitivity-oriented prompt improved recall but still missed 68 of 108 fracture-positive images.

imaging, because the VLM was not fine-tuned and only one multimodal model family was tested. Instead, the result shows that a general-purpose zero-shot model can behave very conservatively on specialized X-ray images, and that sensitivity-oriented prompting improves but does not solve the missed-fracture problem in this setting.

6 Grad-CAM Explainability

Grad-CAM was applied to the best ResNet50 checkpoint to inspect model attention. For a target class c , Grad-CAM computes weights for each feature map A^k in the target convolutional layer by globally averaging gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}. \quad (7)$$

The class-discriminative heatmap is then computed as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right). \quad (8)$$

The heatmap highlights image regions that positively contributed to the model prediction. Since the final feature map is spatially coarse, Grad-CAM should be interpreted as approximate model attention rather than exact fracture localization.

The true positive example shows attention around a musculoskeletal region that plausibly contributed to the fracture prediction. The false positive example shows why explainability is useful for model auditing: the model assigned high fracture probability to a no-fracture image, and the heatmap suggests

attention to a confusing high-contrast anatomical region. The false negative example is especially important because the model assigned low fracture probability to a fracture-positive image. These cases demonstrate that Grad-CAM can reveal useful failure patterns, but it does not prove that the model reasons like a clinician.

In this study, Grad-CAM has a dual role. First, it is used as a qualitative sanity-check tool to inspect whether the model sometimes attends to reasonable anatomical regions. Second, it is used as a failure-analysis tool to identify possible shortcut-learning patterns, such as attention to fixation hardware, image borders, or unrelated high-contrast structures. These two uses are not contradictory, but Grad-CAM should not be treated as evidence of clinical reliability or as proof that the model has identified the true fracture site. It only provides an approximate visualization of spatial evidence used by the network.

7 Discussion

The results support the hypothesis that transfer learning improves fracture classification on this internal FracAtlas split. The custom CNN baseline had lower precision, recall, F1-score, and ROC-AUC than all three pretrained models. MobileNetV2 performed well despite being lightweight, while ResNet50 achieved the best overall balance. DenseNet121 was more conservative, with the highest specificity and lowest false-positive count, but lower recall than ResNet50. Importantly, this conclusion is limited to the FracAtlas distribution used in this experiment; whether the same ranking generalizes to other imaging devices, patient populations, acquisition protocols, or institutions requires

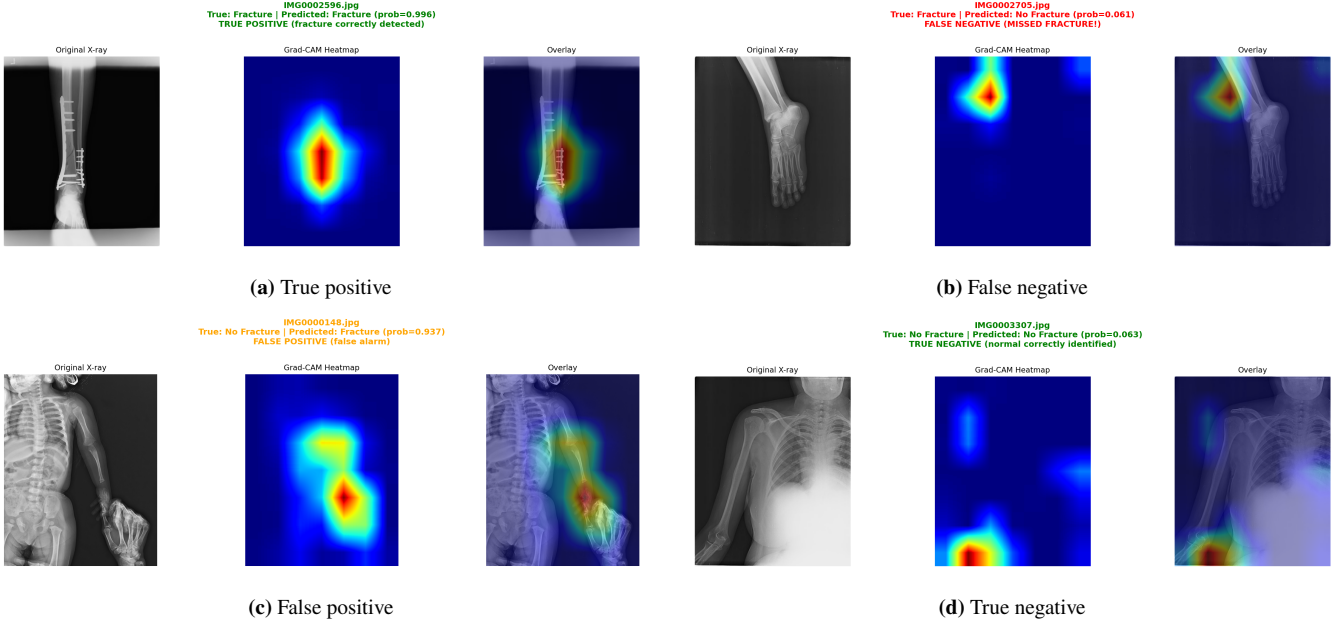


Figure 4: Grad-CAM examples from the ResNet50 model. Correct fracture predictions can focus on musculoskeletal regions, but attention on fixation hardware, image borders, or unrelated high-contrast structures may indicate possible shortcut learning.

external validation.

The most important CNN result is ResNet50’s recall of 0.7593 on the held-out FracAtlas test split. This means the model detected about three-quarters of fracture-positive test images, but it still missed 26 fracture cases. In a medical context, false negatives are more concerning than false positives because they could delay treatment. Therefore, this model is not ready for clinical use. It is better understood as a research prototype showing that pretrained CNNs can learn useful signals within FracAtlas while still requiring stronger validation and safety analysis.

Relative to prior FracAtlas work, the ResNet50 result is promising but should be interpreted cautiously. FAD-MIL reported an AUC of 0.833 and an F1-score of 0.541 on a FracAtlas internal test set under a weakly supervised multiple-instance learning framework [4]. The ResNet50 model in this study achieved a higher ROC-AUC of 0.8915 and F1-score of 0.6534 on the present held-out test split. However, this is not a strict apples-to-apples comparison: FAD-MIL used duplicate removal, tile-based multiple-instance learning, its own split and preprocessing pipeline, and a validation-defined operating threshold, whereas this study used image-level CNN transfer learning and a fixed 0.5 threshold. Therefore, the comparison is best viewed as benchmark positioning rather than evidence of a new state-of-the-art result.

The full-test VLM baseline provides a second finding. GPT-4o-mini could process the same 613 X-ray test images and return structured JSON predictions, but it showed a conservative prediction pattern. The simple and conservative prompts had very high specificity but very low sensitivity, meaning the model rarely produced false alarms but missed most fracture-positive cases. The sensitivity-oriented prompt improved recall from 0.1389 to 0.3704, but it still missed 68 fracture-positive images

and produced 51 false positives. This suggests that prompt wording can shift the VLM’s sensitivity-specificity trade-off, but prompting alone did not produce CNN-level sensitivity in this experiment.

This comparison should be interpreted carefully. The CNN models were trained on FracAtlas, while GPT-4o-mini was evaluated zero-shot without task-specific fine-tuning. Therefore, the result is not a statement that CNNs are always better than VLMs for medical imaging. A more precise interpretation is that, under this evaluation protocol, the task-trained CNN was more sensitive to fractures than the tested general-purpose VLM, especially when false negatives are the main safety concern.

8 Limitations

This project has several limitations. First, the CNN task uses image-level fracture labels rather than bounding-box or segmentation supervision during training. Second, the dataset is class-imbalanced, and weighted loss only partially addresses this problem. Third, the models were trained for a small number of epochs, and validation curves suggest possible overfitting. Fourth, the reported test set comes from the same dataset distribution as the training set; external validation is still needed. Fifth, Grad-CAM heatmaps are low-resolution and cannot be treated as exact fracture localization. Sixth, the model may learn shortcuts from surgical hardware, image quality, positioning, or body-region imbalance rather than fracture morphology alone. Seventh, checkpoint selection used validation F1-score rather than recall alone, which may not select the highest-sensitivity model for a screening-oriented clinical objective. Eighth, the CNN experiments used a fixed 0.5 threshold even though weighted loss can shift the score distribution, so the

reported sensitivity and specificity may not represent the best operating point. Ninth, the custom CNN and transfer-learning models used different epoch counts, which was motivated by different convergence behavior but remains a potential confounding variable. Tenth, the comparison with prior FracAtlas work is indirect because there is not yet a fully standardized public train/validation/test split and evaluation protocol used by all papers.

The VLM baseline also has limitations. It evaluated only GPT-4o-mini, not a broad set of multimodal models. The prompts were limited to three manually designed modes, and each image was queried once per prompt mode, so the experiment does not measure output stability across repeated calls. The returned confidence score is not a calibrated medical probability, even though it can be used for a rough ROC-AUC analysis. The short natural-language reasons were not validated by radiologists. Finally, the OpenAI image-input documentation warns that vision models are not suitable for specialized medical image interpretation or medical advice [14]. For these reasons, the VLM experiment should be treated as a research-only benchmark, not as clinical evidence.

9 Future Work

Future work should explore threshold tuning to prioritize sensitivity, because a lower decision threshold could catch more fractures at the cost of more false positives. Because the CNNs were trained with a positive class weight, threshold selection should be performed systematically on the validation set rather than assuming that 0.5 is the best operating point. For example, future experiments could search thresholds that maximize recall under a minimum precision constraint, or maximize validation F1-score depending on the intended screening objective. Focal loss could be tested as an alternative to weighted binary cross entropy for imbalanced or difficult examples [16]. Probability calibration, including temperature scaling, could be used to make predicted probabilities more reliable [17]. External validation on datasets such as MURA and GRAZPEDWRI-DX would provide a stronger test of generalization across imaging sources and anatomical distributions [2], [3].

For the VLM direction, future work should evaluate stronger multimodal models, medical-domain VLMs if available, and repeated prompt trials. A better VLM evaluation should also include prompt robustness, calibration, uncertainty, image-quality sensitivity, and radiologist review of the generated explanations. Another useful direction is to combine CNN probability outputs with VLM-generated textual explanations, while keeping the classification decision grounded in a validated vision model.

10 Conclusion

This study developed and evaluated a research prototype for fracture identification in X-ray images. Within the internal FracAtlas train/validation/test split, transfer-learning CNNs outperformed the custom CNN baseline, with ResNet50 achieving the best test performance among the evaluated CNNs. Its ROC-AUC

and F1-score are numerically competitive with recent FracAtlas work, although this should be treated as contextual benchmark positioning rather than a strict leaderboard claim because the protocols differ. Grad-CAM visualizations provided a qualitative way to inspect whether the model sometimes focused on relevant musculoskeletal regions, but they also revealed failure cases where attention can shift toward artifacts, hardware, or unrelated anatomy. The full 613-image VLM baseline made the project more current by testing a general-purpose zero-shot multimodal model on the same held-out test set. GPT-4o-mini showed high specificity but low recall, and the sensitivity-oriented prompt improved recall while increasing false positives. Overall, the project demonstrates the value of task-specific CNN transfer learning and qualitative error analysis within this dataset, while emphasizing the need for threshold tuning, external validation, stronger VLM evaluation, and clinical safety analysis before any patient-care use.

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